DATA STRUCTURES AND ALGORITHMS

SAMPLE EXAMINATION PAPER

Course: Data Structures and Algorithms
Time Allowed: 1 Hour 30 Minutes
Total Marks: 50 Marks

Instructions:

Answer all questions.

Write clearly and show your working where necessary.

Draw diagrams where required.

1. SECTION A: SHORT-ANSWER QUESTIONS (10 MARKS)
   a. Define the term *data structure*. (2 marks)
   b. State two differences between a stack and a queue. (2 marks)
   c. What is an algorithm? (2 marks)
   d. Give two examples of linear data structures. (2 marks)
   e. What is the purpose of searching in a data structure? (2 marks)

2. SECTION B: DATA STRUCTURES (20 MARKS)
   a. Arrays and Linked Lists (5 marks)
      i. Define an array. (2 marks)
      ii. State three differences between an array and a linked list. (3 marks)
   b. Stacks and Queues (5 marks)
      i. Explain the LIFO principle used by a stack. (2 marks)
      ii. A stack contains the elements 10, 20, and 30, where 30 is at the top. If the POP operation is performed, what element is removed? (1 mark)
      iii. Explain the FIFO principle used by a queue. (2 marks)
   c. Binary Search Trees (5 marks)
      Insert the following values into an empty Binary Search Tree (BST): 40, 20, 60, 10, 30
      i. Draw the resulting binary search tree. (3 marks)
      ii. State the left child of the root node. (1 mark)
      iii. State the right child of the root node. (1 mark)
   d. Searching and Sorting (5 marks)
      i. Explain the difference between linear search and binary search. (3 marks)
      ii. Arrange the following numbers in ascending order using bubble sort: 8, 3, 6, 2 (2 marks)

3. ALGORITHMS AND COMPLEXITY (20 MARKS)
    a. Algorithms (5 marks)
        i. State three characteristics of a good algorithm. (3 marks)
        ii. Write an algorithm or pseudocode to find the sum of two numbers. (2 marks)
    b. Searching Algorithm (5 marks)
        Consider the sorted array: [5, 10, 15, 20, 25, 30, 35]
        Using binary search:
            i. Identify the middle element in the first search step. (2 marks)
            ii. Explain how the algorithm searches for the value 30. (3 marks)
    c. Time Complexity (5 marks)
        i. What is time complexity? (2 marks)
        ii. What does Big O notation represent? (1 mark)
        iii. State the average or worst-case time complexity of linear search and binary search. (2 marks)
    d. Practical Applications (5 marks)
        i. Give one real-world application of a stack. (1 mark)
        ii. Give one real-world application of a queue. (1 mark)
        iii. Explain why data structures are important in computer programming. (3 marks)